



JYOTI YADAV

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GitHub — LeetCode

EDUCATION

B.Tech – CSE (Data Science)	2023–2027	Netaji Subhas University of Technology (NSUT), New Delhi	8.26/10
AISSCE (Class XII)	2021	Govt Sarvodaya Co-ed Sr Sec School, New Delhi	94.5%
AISCE (Class X)	2019	Ring Midways Public School, New Delhi	88.89%

PROJECTS

E-Commerce Microservices Platform

2026

Spring Boot, Spring Cloud, Kafka, PostgreSQL, Docker, Resilience4j

- Designed a 5-service microservices system (Order, Inventory, Notification, API Gateway, Eureka service discovery) with database-per-service isolation and dynamic load-balanced routing.
- Implemented resilience patterns (circuit breaker, bounded retry, timeouts) on synchronous inter-service calls using Resilience4j, with selective retry logic that excludes business-rule failures.
- Built a saga pattern with compensating transactions to maintain consistency across services without distributed transactions.
- Ensured safe concurrent stock updates using idempotency keys and optimistic locking; handled Kafka consumer failures via dead-letter topics.

Event Management System

2026

React.js, Spring Boot, Spring Security, JWT, JPA/Hibernate, MySQL, REST APIs

- Built a full-stack event management platform with JWT-based authentication, a custom filter chain, and role-based route protection for Organizer and Attendee roles.
- Designed a layered REST API (Controller–Service–Repository) with a custom exception hierarchy and centralized global exception handling.
- Modeled a normalized MySQL schema (User, Event, Registration) enforcing business rules such as duplicate registration checks and capacity limits at the service layer.
- Built a responsive React frontend with protected routes, registration workflow, and Axios-based API integration.

Hierarchical Vision Transformer Ensemble for Knee Osteoarthritis Diagnosis

Mar 2026

Python, PyTorch, OpenCV, Scikit-learn

- Built an ensemble-based image classification pipeline combining Swin Transformer, MaxViT, and GCViT models for 5-class knee osteoarthritis grading.
- Developed an image preprocessing pipeline using CLAHE, Gaussian filtering, and Laplacian enhancement.
- Addressed class imbalance using WeightedRandomSampler during model training.
- Achieved an **AUC of 0.911** on the evaluation dataset (71.14% accuracy across 5 classes).

POSITIONS OF RESPONSIBILITY

Team Lead – Event Management System (DBMS Project)

Led a team of 3 members, designing the ER diagram, finalizing the relational schema, coordinating implementation, and managing integration testing.

Team Lead – Fake News Detection (ML Project)

Led a team of 3 members, coordinating dataset preprocessing, feature engineering, model evaluation, and experimental reporting.

EXTRA-CURRICULAR ACTIVITIES AND ACHIEVEMENTS

Community Service Intern – Nanhi Kashtiyani NGO

May 2024 – Jun 2024

Volunteered as an English tutor for underprivileged children, strengthening communication and mentoring skills.

Solved **270+** DSA problems (Java) on LeetCode, strengthening analytical thinking and coding proficiency.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript

Frameworks & Technologies: Spring Boot, Spring Security, Spring Data JPA, Hibernate, React.js, REST APIs, JWT

Databases: MySQL, PostgreSQL

Developer Tools: Git, GitHub, Docker, Maven, Postman, Swagger/OpenAPI, IntelliJ IDEA, VS Code

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks